

Multiply de re counterpart theory and actuality

Handout for *Time and Modality* seminar

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1 A bad idea

$$\begin{aligned}\text{BMCT+} \quad \Diamond\phi(t_1, \dots, t_n) &:= \exists v_1 \dots \exists v_n (Cv_1 t_1 \wedge \dots \wedge Cv_n t_n \wedge \phi(x_1, \dots, x_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \forall v_1 \dots \forall v_n ((Cv_1 t_1 \wedge \dots \wedge Cv_n t_n) \rightarrow \phi(x_1, \dots, x_n))\end{aligned}$$

Basic problem: too easy for a possibility claim to be true, too hard for a necessity claim to be true.

Illustrations:

- (i) Possibly, Socrates is not a member of {Socrates}.
- (ii) The proposition that Socrates is wise is such that possibly, Socrates is wise and it is false.
- (iii) Possibly, the Continental US is not part of the US.
- (iv) Many people are possibly mothers of Charles.

Problems for temporal analogue:

- I have not always lived on Earth.
- Many people are sometimes mothers of Charles.

2 A bad fix

$$\begin{aligned}\text{LCT+} \quad \Diamond\phi(t_1, \dots, t_n) &:= \exists w v_1 \dots v_n (Ww \wedge Iv_1 w \wedge \dots \wedge Iv_n w \\ &\quad \wedge Cv_1 t_1 \wedge \dots \wedge Cv_n t_n \wedge \phi(v_1, \dots, v_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \forall w v_1 \dots v_n ((Ww \wedge Iv_1 w \wedge \dots \wedge Iv_n w \\ &\quad \wedge Cv_1 t_1 \wedge \dots \wedge Cv_n t_n) \rightarrow \phi(v_1, \dots, v_n))\end{aligned}$$

- Old problems not solved if things can have multiple counterparts in the same Lewis-world.
- Old problem (iv) (involving unrestricted quantification) not solved even if we require that everything have at most one counterpart per Lewis-world.
- New problem: failure of K.
- New problem: failure of T.

3 Lewis's fix: counterparts of sequences

$$\begin{aligned} \text{SEQUENCES} \quad \Diamond\phi(t_1, \dots, t_n) &:= \exists v_1 \dots \exists v_n (C\langle v_1, \dots, v_n \rangle \langle t_1, \dots, t_n \rangle \wedge \phi(v_1, \dots, v_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \forall v_1 \dots \forall v_n (C\langle v_1, \dots, v_n \rangle \langle t_1, \dots, t_n \rangle \rightarrow \phi(v_1, \dots, v_n)) \end{aligned}$$

Constraints on the sequence-counterpart relation are needed, for example:

Permutation: If π is a function from the natural numbers $< m$ to the natural numbers $< n$, and σ_1 and σ_2 are sequences of length n such that σ_2 is a counterpart of σ_1 , then $\sigma_2 \circ \pi$ is a counterpart of $\sigma_1 \circ \pi$.

4 Counterpairings

$$\begin{aligned} \text{COUNTERPAIRINGS} \quad \Diamond\phi(t_1, \dots, t_n) &:= \exists R \exists v_1 \dots \exists v_n (\mathbf{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n \wedge \phi(v_1, \dots, v_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \forall R \forall v_1 \dots \forall v_n ((\mathbf{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n) \rightarrow \phi(v_1, \dots, v_n)) \end{aligned}$$

If we require counterpairings to be *functional* $((\mathbf{CR} \wedge Rxy \wedge Rxz) \rightarrow y = z)$ and to have *universal domains* $(\mathbf{CR} \rightarrow \forall x \exists y Rxy)$, rewrite more compactly as follows:

$$\begin{aligned} \text{COUNTERPAIRINGS}^* \quad \Diamond\phi(t_1, \dots, t_n) &:= \exists R (\mathbf{CR} \wedge \phi(R[t_1], \dots, R[t_n])) \\ \Box\phi(t_1, \dots, t_n) &:= \forall R (\mathbf{CR} \rightarrow \phi(R[t_1], \dots, R[t_n])) \end{aligned}$$

Extending to plurals:

$$\begin{aligned} Rxx \ yy &:= \forall x (x \in xx \rightarrow \exists y (y \in yy \wedge Rxy)) \wedge \\ &\quad \forall y (y \in yy \rightarrow \exists x (x \in xx \wedge Rxy)) \end{aligned}$$

Extending to higher-order terms: Easy to begin:

$$\begin{aligned} RPQ &:= (\text{Qualitative}(P) \wedge Q = P) \vee \\ &\quad \exists F \exists x \exists y (\text{Qualitative}(F) \wedge P = Fx \wedge Q = Fy \wedge Rxy) \vee \\ &\quad \exists R \exists x_1 \exists x_2 \exists y_1 \exists y_2 (\text{Qualitative}(R) \wedge P = Rx_1 x_2 \wedge Q = Ry_1 y_2 \wedge Rx_1 y_1 \wedge Rx_2 y_2) \vee \\ &\quad \vdots \end{aligned}$$

But how to deal with the infinite case?

5 Constraints corresponding to modal axioms

<i>Axiom</i>	<i>Constraint</i>
Duality $\Box\neg\phi \leftrightarrow \neg\Diamond\phi$	Automatic
K $\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$	Counterpairings have universal domains: $\mathbf{CR} \rightarrow \forall x\exists yRxy$
D $\Box\phi \rightarrow \Diamond\phi$	There is at least one counterpairing: $\exists RCR$
T $\Box\phi \rightarrow \phi$	The identity relation is a counterpairing: $\forall x\forall y(Rxy \leftrightarrow x = y) \rightarrow \mathbf{CR}$
B $\phi \rightarrow \Box\Diamond\phi$	R^{-1} (the converse of R) is a counterpairing when R is: $(\mathbf{CR} \wedge \forall x\forall y(Sxy \leftrightarrow Ryx)) \rightarrow \mathbf{CS}$
4 $\Box\phi \rightarrow \Box\Box\phi$	$R \circ S$ (the relative product of R with S) is a counterpairing when R and S are: $(\mathbf{CR} \wedge \mathbf{CS} \wedge \forall x\forall y(Qxy \leftrightarrow \exists z(Sxz \wedge Rzy))) \rightarrow \mathbf{CQ}$
5 $\Diamond\phi \rightarrow \Box\Diamond\phi$	$R \circ S^{-1}$ is a counterpairing when R and S are: $(\mathbf{CR} \wedge \mathbf{CS} \wedge \forall x\forall y(Qxy \leftrightarrow \exists z(Szx \wedge Rzy))) \rightarrow \mathbf{CQ}$
NI $x = y \rightarrow \Box x = y$ NO $Xx \rightarrow \Box Xx$	Counterpairings are functional: $(\mathbf{CR} \wedge Rxy \wedge Rxz) \rightarrow y = z$
ND $x \neq y \rightarrow \Box x \neq y$ NNO $\neg Xx \rightarrow \Box \neg Xx$	
CBF $\Box\forall x\phi \rightarrow \forall x\Box\phi$ CBF2 $\Box\forall X\phi \rightarrow \forall X\Box\phi$	Automatic
BF $\forall x\Box\phi \rightarrow \Box\forall x\phi$	Every counterpairing can be extended so that any given object is in its range: $\mathbf{CR} \rightarrow \forall x\exists S(\mathbf{CS} \wedge \forall z_1\forall z_2(Rz_1z_2 \rightarrow Sz_1z_2) \wedge \exists yS yx)$
BF2 $\forall X\Box\phi \rightarrow \Box\forall X\phi$	Counterpairings have universal ranges: $\mathbf{CR} \rightarrow \forall y\exists xRxy$

6 Constraints corresponding to tense-logical axioms

<i>Axiom</i>	<i>Constraint</i>
Mixing _P $\phi \rightarrow \mathbf{HF}\phi$	R is a past-counterpairing $\rightarrow R^{-1}$ is a future-counterpairing
Mixing _F $\phi \rightarrow \mathbf{GP}\phi$	R is a future-counterpairing $\rightarrow R^{-1}$ is a past-counterpairing
Totality _P $(\mathbf{P}\phi \wedge \mathbf{P}\psi) \rightarrow (\mathbf{P}(\phi \wedge \mathbf{P}\psi) \vee \mathbf{P}(\phi \wedge \psi) \vee \mathbf{P}(\psi \wedge \mathbf{P}\phi))$	If R and S are past-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a past-counterpairing, or $R = S$.
Totality _F $(\mathbf{F}\phi \wedge \mathbf{F}\psi) \rightarrow (\mathbf{F}(\phi \wedge \mathbf{F}\psi) \vee \mathbf{F}(\phi \wedge \psi) \vee \mathbf{F}(\psi \wedge \mathbf{F}\phi))$	If R and S are future-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a future-counterpairing, or $R = S$.
Density _P $\mathbf{P}\phi \rightarrow \mathbf{PP}\phi$	If R is a past-counterpairing, then $R = S \circ T$ for some past-counterpairings S and T
Density _F $\mathbf{F}\phi \rightarrow \mathbf{FF}\phi$	If R is a future-counterpairing, then $R = S \circ T$ for some future-counterpairings S and T

7 Williamson and Fara on ‘actually’

- (L1) $[\text{ACT } \phi(a)]^v$ is $\exists x(Ix@ \wedge Cxa \wedge [\phi(x)]^v)$
 (L2) $[\text{ACT } \phi(a)]^v$ is $\forall x((Ix@ \wedge Cxa) \rightarrow [\phi(x)]^v)$
 (L1') $[\text{ACT } \phi(a)]^v$ is $\exists x(Ix@ \wedge Cax \wedge [\phi(x)]^v)$
 (L2') $[\text{ACT } \phi(a)]^v$ is $\forall x((Ix@ \wedge Cax) \rightarrow [\phi(x)]^v)$

All have a problem with (12):

$$(12) \quad \Diamond \exists x(\text{ACT } Fx \leftrightarrow \text{ACT } \neg Fx)$$

Also worth thinking about:

- (*) $\forall x(Fx \rightarrow \Box \text{ACT } Fx)$
 (**) $\forall x(\Diamond \text{ACT } Fx \rightarrow Fx)$

8 ‘Actually’ with counterpairings

Require counterpairings to be *global permutations* (functional, one-one, universal domains, universal ranges). Then we can “follow them back” when we need to.

Sample translations:

$$\begin{array}{ll} \exists x \Diamond \exists y (F(x, y) \wedge \text{ACT } G(x, y)) & \exists x \exists R (\mathbf{C}R \wedge \exists y (F(R[x], y) \wedge G(x, R^{-1}[y]))) \\ \Diamond \Diamond \exists x (Fx \wedge \text{ACT } Gx) & \exists R (\mathbf{C}R \wedge \exists S (\mathbf{C}S \wedge \exists x (Fx \wedge G(S^{-1}[R^{-1}[x]])))) \end{array}$$

How to implement?